Localized Corrosion of Molybdenum-Bearing Nickel Alloys in Chloride Solutions☆

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Abstract

Electrochemical and immersion tests have been applied to a study of the localized corrosion resistance of two molybdenum-bearing nickel alloys, Alloys C-276 and 625, in neutral chloride solutions in the temperature range of 25 to 200 C as part of the container materials evaluation screening tests for the Canadian Nuclear Fuel Waste Management Program. Cyclic polarization studies show that the passivation breakdown potentials move rapidly to more active values with increasing temperatures, indicating a reduced resistance to localized corrosion. The results of immersion tests show that both alloys do suffer crevice corrosion in neutral aerated sodium chloride solutions at elevated temperatures, but that in both cases, there is a limiting temperature >100 C, below which, the alloys are not attacked, regardless of the chloride concentration.

Introduction

Two molybdenum-bearing nickel alloys, Alloys C-276 and 625, were included in the group of passive alloys identified by Nuttall and Urbanic[1] as possible candidate container materials for the Canadian Nuclear Fuel Waste Management Program. The concept being assessed involves the disposal of nuclear waste by deep burial in the Canadian Shield. The deep groundwaters of the Canadian Shield may have a high chloride content,[2] and the possibility of passivation breakdown and subsequent localized corrosion (crevice and/or pitting) of any passive metals considered for the containers must be considered.

The assessment of materials by Nuttall and Urbanic[1] revealed a paucity of data for the localized corrosion of nickel alloys in the range of temperature and chloride content expected in a disposal vault. At that time, it was assumed for the purpose of assessment "that the container temperature will be 150 C or less." However, more recently, the maximum container-skin temperature specified[3] for the concept assessment engineering studies is 100 C.

It is likely that oxygen will be present initially as a result of the burial operations and, subsequently, as a result of radiolysis.

The present study, which involved electrochemical and immersion tests, was conducted to determine the conditions (temperature and chloride content) required to stimulate localized corrosion of Alloys C-276 and 625. It was decided to use a wide range of temperatures, 25 to 200 C, and NaCl concentrations of 0.1 wt% to saturation so that the results may assist in the understanding of the mechanism of the localized corrosion, rather than simply obtain data for specific concept assessment guidelines.

An understanding of the mechanism would be required to construct the mathematical models necessary to extrapolate the results to very long exposure times, as discussed later.

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Experimental

Test Specimens

The test specimens were machined from plate of a composition shown in Table 1 in the as received, mill-annealed condition. The surfaces were wet ground on silicon carbide papers down to 600 grade.

Two types of specimens were used:

1. Plain blocks, 11 x 11 x 7 mm thick, with a drilled and tapped connection, used as electrodes with a polytetrafluorethylene (PTFE) compression gasket and a PTFE insulated rod made from the same alloy.

2. Blocks, 25 x 13 x 7 mm thick, fitted with a radially grooved 16-mm diameter PTFE washer assembly bolted to the block with a bolt made from the alloy under test. A plain PTFE backing washer 16-mm diameter was used. A torque of 6 Nm was used. Some of these specimens had a drilled and tapped connection and were used as electrodes; the others were used for the immersion tests.

Apparatus

Conventional multinecked 1-L glass corrosion cells[4] were used at temperatures ≤100 C for both the electrochemical and immersion tests. An external SCE with a solution bridge and Luggin probe were used for the electrochemical measurements.

The exposure tests at t > 100 and ≤ 150 C were conducted in closed tubular glass cells with an air space heated in an oil bath. At t > 150 C, similar glass cells were heated in a Parr 1-L stainless steel (SS) autoclave containing water.

The electrochemical studies at t > 100 C were performed in a Parr 1-L titanium autoclave fitted with an external Ag/AgCl reference electrode that was maintained at 25 C and system pressure.[5]

An EG & G[(1)] PARC 350A, electrochemical corrosion system, with IR compensation, was used for the cyclic polarization localized corrosion studies.

Procedures

The cyclic potentiodynamic polarization tests were conducted in solutions deaerated with nitrogen, starting at E_{corr} with dE/dt = 20 mV/min. The scans were reversed at I = 1 mA/cm² and were

[(1)]EG & G Princeton Applied Research, Princeton, New Jersey.

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TABLE 1 — Ni-Alloy Analyses

Alloy	Composition (wt%) Ni	Cr	Mo	Fe	C	Mn	Si	S	Cb + Ta	P	V	W	Co	Al	Ti	UNS No.
C-276	bal	15.70	15.69	6.06	0.002	0.47	0.02	0.002	—	0.008	0.14	3.65	1.87	—	—	N10276
625	60.80	21.63	8.52	4.68	0.03	0.10	0.31	0.001	3.54	0.009	—	—	—	0.14	0.24	N06625

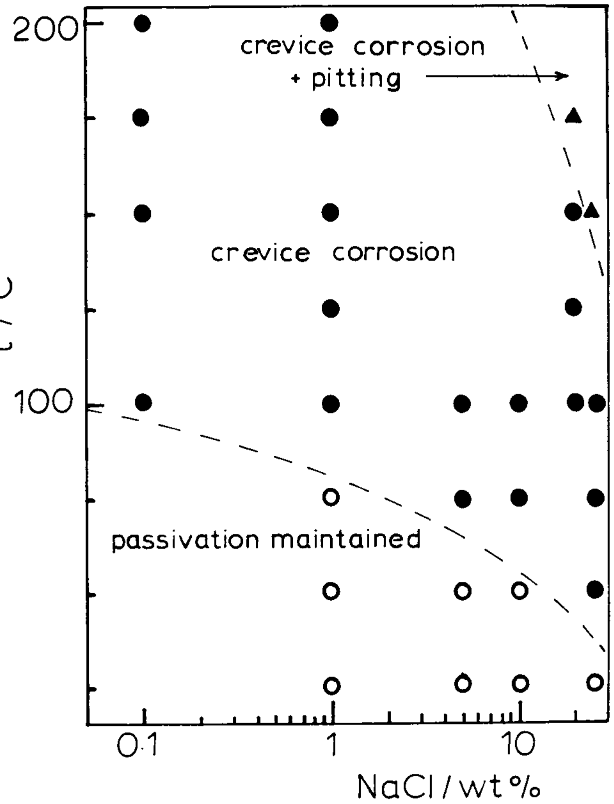

FIGURE 1 — The nature of the localized corrosion obtained by cyclic polarization with Alloy C-276.

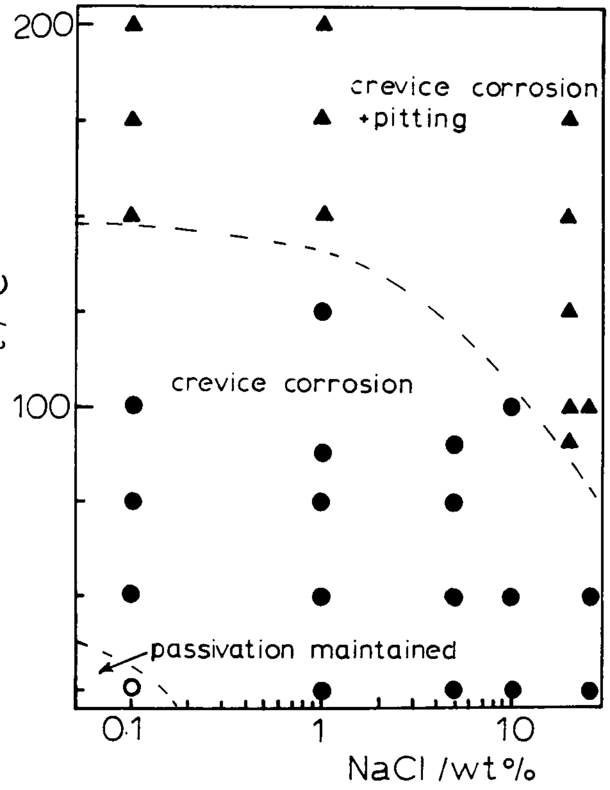

FIGURE 2 — The nature of the localized corrosion obtained by cyclic polarization with Alloy 625.

terminated below the repassivation potential.

The exposure tests were conducted in continuously aerated solutions at t ≤ 100 C and in closed cells with an air space at t > 100 C.

The specimens were subsequently examined for corrosion with a stereomicroscope to x 160. Some specimens were examined by scanning electron microscopy (SEM).

Results and Discussion

Electrochemical Studies

The cyclic potentiodynamic polarization method was used to determine the susceptibility of the alloys to localized corrosion. Both plain and creviced specimens were used in the initial experiments. It was subsequently found unnecessary to use a crevice assembly in the electrochemical tests, since in every case where localized corrosion occurred at the artificial crevice assembly, it also occurred at the PTFE compression gasket located at the electrode mount. Perhaps, it is not surprising that Asphahani[6] coined the term *damaging potential* to describe passivation breakdown potentials rather than using the terms pitting potentials or crevice potentials since it is difficult to distinguish between these two forms of attack with cyclic polarization tests. Indeed, the round robin tests[7] conducted in conjunction with the standard cyclic polarization test, ASTM[(2)] Standard G-61, illustrated this fact very well, and the standard test makes no attempt to distinguish between pitting and crevice corrosion.

The three conditions that were observed (Figures 1 and 2) when the electrodes were subsequently examined were passivation maintained with no corrosion, crevice corrosion at the PTFE electrode mount, and crevice corrosion at the PTFE electrode mount plus pitting on the freely exposed surfaces. On this basis, Alloy C-276 displayed superior resistance to localized corrosion than Alloy 625. The cyclic polarization test for localized corrosion is a severe test involving the polarization of a metal to potentials that may not be achieved under natural corrosion conditions.

At high temperatures, the alloys exhibited classical hysteresis loops [Figures 3(a) and (b)], characteristic of passivation breakdown on the upward sweep and repassivation on the downward sweep. Under these conditions, passivation breakdown potentials, E_b, and repassivation potentials (sometimes called the protection potential), E_p could be readily obtained from the polarization curves. The conversion of the potentials to the SHE(T) scale was achieved using the method given by Macdonald;[8] the thermal junction potentials were assumed to be negligible.

At lower temperatures, the polarization curves were more complex and the following characteristics, which made determination of E_b and E_p difficult, were observed.

1. The absence of a hysteresis loop [Figure 3(c)] under conditions where localized attack was found on subsequent examination of the specimen.

2. The presence of two transpassive zones [Figure 3(d)], the lower probably corresponding to the anodic oxidation of molybdenum and the upper to chromium.[9]

[(2)]American Society for Testing and Materials (ASTM), Philadelphia, Pennsylvania.

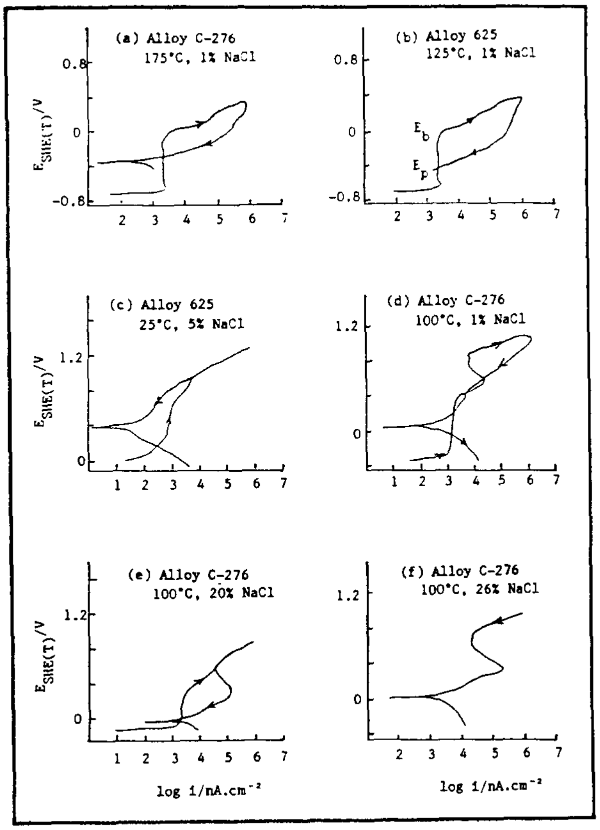

FIGURE 3 — Cyclic polarization scans.

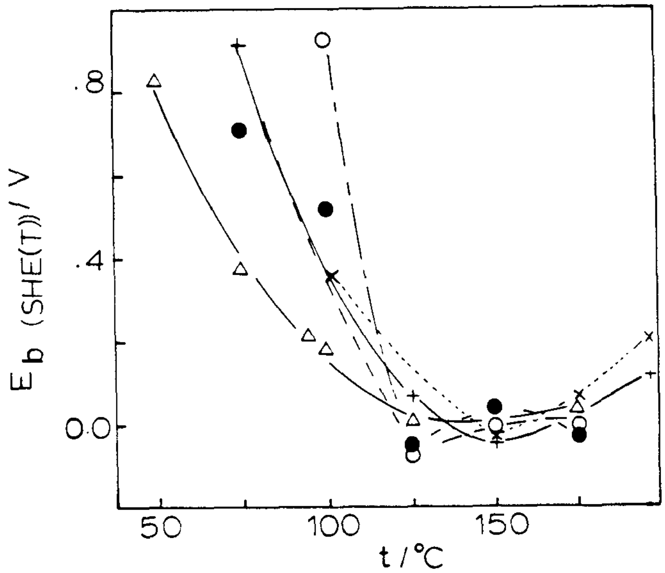

FIGURE 4 — Passivation breakdown potentials; (+) Alloy 625, 0.1% NaCl, (Δ) Alloy 625, 20% NaCl, (•) Alloy C-276, 20% NaCl, (o) Alloy C-276, 1% NaCl, and (X) Alloy C-276, 0.1% NaCl.

3. The presence of reversible behavior on the chromium dissolution zone on the downward sweep with a hysteresis loop occurring below this reversible section [Figure 3(e)]. The reversible behavior on the downward sweep was similar when the sweep was interrupted, and the potential was held at the most noble value for a period of 4 h [Figure 3(f)], ruling out a simple time effect.

Not withstanding the difficulties of interpreting the electrolysis curves at the lower temperatures, the overall results showed that the resistance to localized corrosion decreased rapidly as the temperature was raised, as evidenced by the decrease in the breakdown potentials to much more active values (Figure 4).

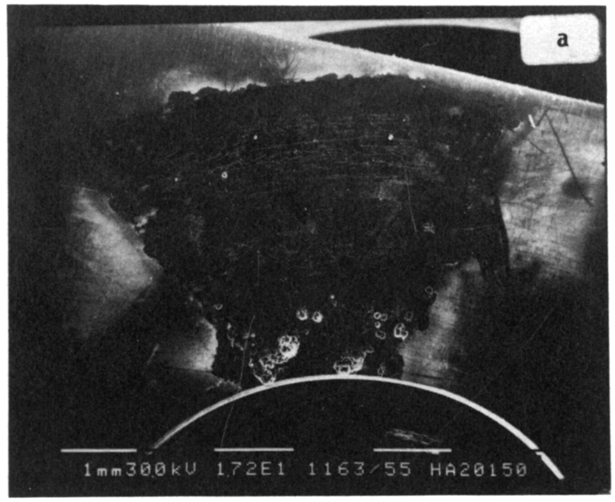

FIGURE 5 — Crevice corrosion following immersion tests with Alloy C-276 in aerated 20% NaCl at 150 C.

In previous studies with nickel,[10] various austenitic SSs,[5,11,12] iron,[5] and a zirconium alloy,[13] a reversal temperature was reported, above which, the trend to more active breakdown potentials with increasing temperature was reversed. In some cases, the reversal led to a greater resistance to pitting at elevated temperatures than at room temperature. Hickling and Wieling[14] reported that the pitting resistance of AISI[(3)] 347 SS and two nickel alloys, Alloys 600 and 800, increased with increasing temperature in the range of 150 to 250 C. The results presented here for Alloys C-276 and 625 show a leveling off of the breakdown potentials above ~125 C, but they do not reveal a significant reversal of the trend, and further work at temperatures up to 300 C would be required to clarify the situation with respect to the presence of a reversal temperature.

Immersion Tests

Corrosion in both alloys took the form of crevice corrosion with no pitting observed outside the crevices. The crevice corrosion (Figure 5) took the form of pitting and was usually found under only one or two of the crevices formed by the PTFE washer or the backing washer.

The results of the immersion tests with creviced specimens are presented in Figures 6 and 7, where it can be seen that, as expected, the crevice corrosion temperature (CCT) is a function of the chloride concentration.

The effect of Cr and Mo additions on the localized corrosion resistance of passive alloys is well known,[15] and the superior

[(3)]American Iron and Steel Institute, Washington, DC.

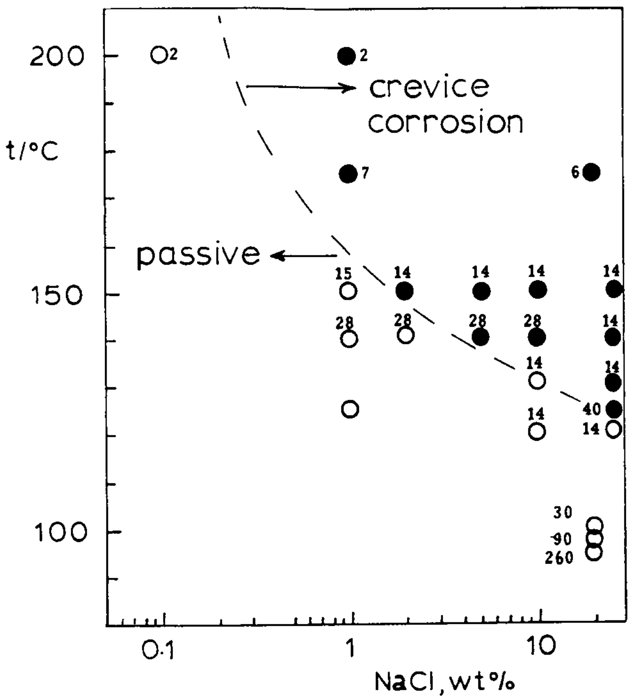

FIGURE 6 — Effect of temperature and chloride concentration on the crevice corrosion of Alloy C-276. The number beside each data point is the exposure time in days; (•) crevice corrosion with obvious metal loss, (o) no corrosion, and (- - - - -) tentative crevice temperature envelope.

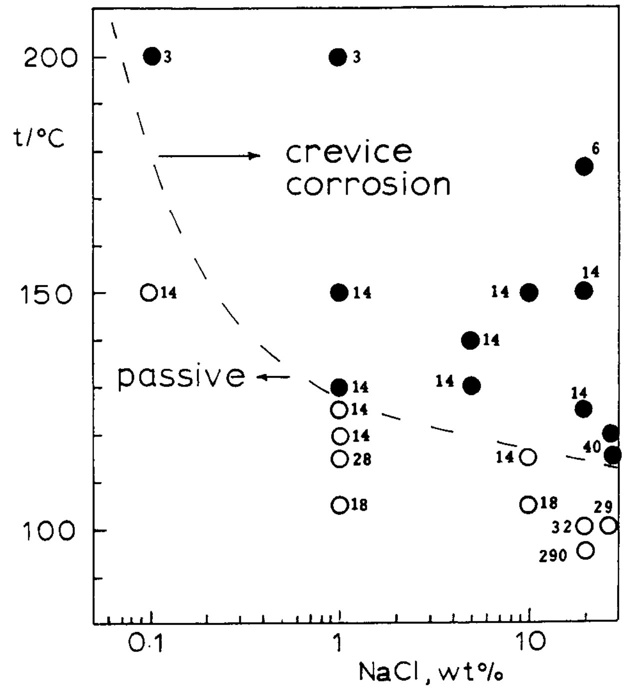

FIGURE 7 — Effect of temperature and chloride concentration on the crevice corrosion of Alloy 625; the number beside each data point is the exposure time in days; (•) crevice corrosion with obvious metal loss, (o) no corrosion, and (- - - - -) tentative crevice temperature envelope.

performance of Alloy C-276 most probably relates to the higher Mo content of the alloy. For example, Rockel has proposed a pitting index for SSs on the basis of the (Cr + 3 x Mo) content.[16]

Both alloys appear to exhibit a limiting CCT between 100 and 125 C, below which, crevice corrosion does not occur, regardless of the chloride concentration. This result is in general agreement with the results of the electrochemical tests, which showed that the breakdown potentials moved rapidly to more active values above 100 C, suggesting the possibility of localized corrosion in aerated solutions where the presence of dissolved oxygen could raise the potential above the breakdown potential.

Szklarska-Smialowska and Mankowski[18] concluded that a reproducible critical potential for crevice corrosion exists that is more negative than the critical potential for pit nucleation and considered that the pitting of the five SSs studied was a special kind of pitting corrosion. If this were the case in the present study, which the morphology of the attack (Figure 5) certainly supports, the agreement between the electrochemical results and the immersion tests for the concentrated chloride solutions could be understood.

The results for the dilute solutions are not so easily explained. The E_b values in the dilute solutions at t = 125 C were similar to those in the concentrated solutions with no systematic variation with chloride concentrations. But corrosion was initiated only at much higher temperatures in the immersion tests with the dilute solutions, and, clearly, factors other than the value of E_b and the presence of large hysteresis loops are involved.

The lack of agreement at the lower chloride concentrations can be explained if the differential aeration cell/acidification theory of crevice corrosion, which was mathematically modeled by Oldfield and Sutton,[19] is applied. The authors stated on the basis of the behavior of this model that "the bulk chloride level is clearly of extreme importance in determining when, if at all, crevice corrosion will occur." Crevice corrosion, on the basis of this theory, starts when the differential aeration cell produces acidification and chloride enhancement in the crevice to the point where a critical crevice solution is produced, which can disrupt the stable passive film. The Oldfield-Sutton model was successfully applied to a study of crevice corrosion of SS in seawater by Kain,[20] where again, the major effect of the bulk concentration of Cl^- is illustrated.

On the basis of the Oldfield-Sutton model, it must be assumed that either the exposure times at the low chloride concentrations were too short or else the critical crevice solution could not be attained at any exposure time at the lower temperatures.

Oldfield and Sutton did not consider either breakdown or protection potentials to be appropriate parameters for inclusion in their crevice corrosion model, although their experimental results[21] with AISI 316 SS showed a potential/time relationship characteristic of passivation breakdown obtained when the breakdown potential theory is applied,[17] albeit at potentials lower than those normally associated with the passivation breakdown of AISI 316 SS. Further the initial stages of the attack were said to take the form of micropits, which only later coalesced to give active dissolution.

Ni Alloys and Nuclear Waste Containers

If the existence of a limiting crevice corrosion temperature for Alloys C-276 and 625 above 100 C is confirmed, the molybdenum-bearing nickel alloys would be strong candidates for further consideration for nuclear waste disposal containers. The significance of this limiting temperature is that possible Cl^- concentration at the canister surface, because of boiling before the full hydrostatic pressure develops, would not cause a problem in terms of localized corrosion.

The recent evaluation of corrosion screening tests conducted for the Commission of the European Communities[22] concluded that the molybdenum-bearing nickel alloys exhibited good resistance to corrosion in chloride solutions, with Alloy C-276 ranked higher than Alloy 625, although the nickel alloy selected for further study, Alloy C-4, did suffer crevice corrosion at 90 C in a $MgCl_2$-rich brine under gamma radiation.

Clearly, the type of data presented here, which was intended to fill a gap in our knowledge and understanding of the localized corrosion behavior of molybdenum-bearing nickel alloys, must be supplemented by more extensive long-term testing. And as pointed out by Accary,[22] further experimental studies should be paralleled by further development of the mathematical modeling of the type developed by Oldfield and Sutton[19,21] and used by Kain.[20] No corrosion model presently exists that would allow for the prediction of localized corrosion over the containment durations involved in waste disposal, especially if long-term factors, such as solid state diffusion leading to aging of alloys and penetration of environmental elements into the alloy, or loss of its components, are to be taken into account.[22]

Not withstanding the difficulties in such modeling, such an approach is required so that medium to long-term (10 to 50 year) pilot tests can be extrapolated to the 500 year desired minimum container effectiveness.

Acknowledgment

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